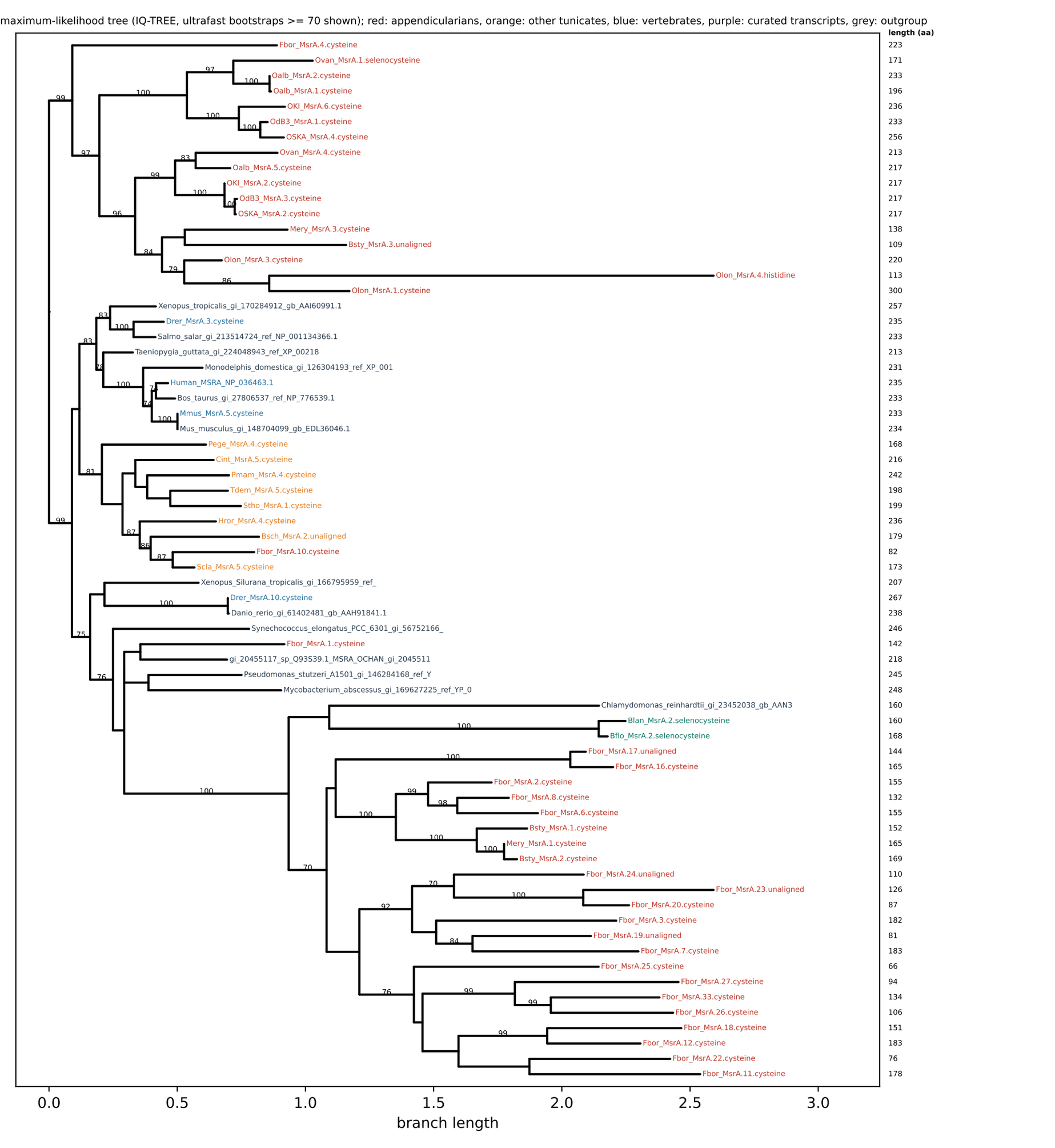
maximum-likelihood tree (IQ-TREE, ultrafast bootstraps >= 70 shown); red: appendicularians, orange: other tunicates, blue: vertebrates, purple: curated transcripts, grey: outgroup
length (aa)
Fbor_MsrA.4.cysteine 223
Ovan_MsrA.1.selenocysteine 171
Oalb_MsrA.2.cysteine 233
Oalb_MsrA.1.cysteine 196
OKI_MsrA.6.cysteine 236
OdB3_MsrA.1.cysteine 233
OSKA_MsrA.4.cysteine 256
Ovan_MsrA.4.cysteine 213
Oalb_MsrA.5.cysteine 217
OKI_MsrA.2.cysteine 217
OdB3_MsrA.3.cysteine 217
OSKA_MsrA.2.cysteine 217
Mery_MsrA.3.cysteine 138
Bsty_MsrA.3.unaligned 109
Olon_MsrA.3.cysteine 220
Olon_MsrA.4.histidine 113
Olon_MsrA.1.cysteine 300
Xenopus_tropicalis_gi_170284912_gb_AAI60991.1 257
Drer_MsrA.3.cysteine 235
Salmo_salar_gi_213514724_ref_NP_001134366.1 233
Taeniopygia_guttata_gi_224048943_ref_XP_00218 213
Monodelphis_domestica_gi_126304193_ref_XP_001 231
Human_MSRA_NP_036463.1 235
Bos_taurus_gi_27806537_ref_NP_776539.1 233
Mmus_MsrA.5.cysteine 233
Mus_musculus_gi_148704099_gb_EDL36046.1 234
Pege_MsrA.4.cysteine 168
Cint_MsrA.5.cysteine 216
Pmam_MsrA.4.cysteine 242
Tdem_MsrA.5.cysteine 198
Stho_MsrA.1.cysteine 199
Hror_MsrA.4.cysteine 236
Bsch_MsrA.2.unaligned 179
Fbor_MsrA.10.cysteine 82
Scla_MsrA.5.cysteine 173
Xenopus_Silurana_tropicalis_gi_166795959_ref_ 207
Drer_MsrA.10.cysteine 267
Danio_rerio_gi_61402481_gb_AAH91841.1 238
Synechococcus_elongatus_PCC_6301_gi_56752166_ 246
Fbor_MsrA.1.cysteine 142
gi_20455117_sp_Q93S39.1_MSRA_OCHAN_gi_2045511 218
Pseudomonas_stutzeri_A1501_gi_146284168_ref_Y 245
Mycobacterium_abscessus_gi_169627225_ref_YP_0 248
Chlamydomonas_reinhardtii_gi_23452038_gb_AAN3 160
Blan_MsrA.2.selenocysteine 160
Bflo_MsrA.2.selenocysteine 168
Fbor_MsrA.17.unaligned 144
Fbor_MsrA.16.cysteine 165
Fbor_MsrA.2.cysteine 155
Fbor_MsrA.8.cysteine 132
Fbor_MsrA.6.cysteine 155
Bsty_MsrA.1.cysteine 152
Mery_MsrA.1.cysteine 165
Bsty_MsrA.2.cysteine 169
Fbor_MsrA.24.unaligned 110
Fbor_MsrA.23.unaligned 126
Fbor_MsrA.20.cysteine 87
Fbor_MsrA.3.cysteine 182
Fbor_MsrA.19.unaligned 81
Fbor_MsrA.7.cysteine 183
Fbor_MsrA.25.cysteine 66
Fbor_MsrA.27.cysteine 94
Fbor_MsrA.33.cysteine 134
Fbor_MsrA.26.cysteine 106
Fbor_MsrA.18.cysteine 151
Fbor_MsrA.12.cysteine 183
Fbor_MsrA.22.cysteine 76
Fbor_MsrA.11.cysteine 178
0.0
0.5
1.0
1.5
2.0
2.5
3.0
branch length